



Yuxuan Liu

Ph.D. Candidate in Trustworthy Generative Audio and Music AI ·
Expected to Graduate Mar. 2027

AI Algorithms · Audio
Intelligence
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Google Scholar

Education

Beijing University of Posts and Telecommunications · 2017– B.Eng. in Internet of Things Engineering	2021	University of Warwick · M.Sc. in Cyber Security Engineering	2021– 2022
Xi'an Jiaotong-Liverpool University · Ph.D. Candidate, School of Intelligent Engineering	2023–2027 (expected)		

Research Profile

Ph.D. candidate at the School of Intelligent Engineering, Xi'an Jiaotong-Liverpool University, focusing on trustworthy generative audio, music intelligence, and AI security. My methodological focus is **privacy auditing, adversarial generation, and perceptually aligned evaluation**. I propose and validate methods for membership inference, music diffusion, and symbolic-music models; as a project lead, I also advance MusicXML-to-audio rendering research into a creator-facing proof of concept.

Research and Project Experience

Innovative methods → creator-facing delivery

2020–2021 · Audio Algorithm Foundations Contributed to singing voice generation and enhancement research at Changba.	2023–2027 · Ph.D. Research Propose and validate generative-audio security methods.	Mar. 2027 · Ph.D. Graduation Available for AI, audio intelligence, and trustworthy-model roles.
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Trustworthy Generative Audio and Music Intelligence Ph.D. Research · Researcher

Role	Membership inference determines whether a work was used in model training, making it important for auditing data-privacy and copyright risks in generative models.
Action	Develop membership-inference and auditing methods for music diffusion and symbolic-music generation models, with emphasis on reliable identification at low false-positive rates.
Outcome	Produced 9 publications and workshop presentations across ICASSP, ISMIR, AAAI Workshop, CMMR, FG, and DCASE.

Music Adversarial Attacks and Perceptual Evaluation Ph.D. Research · First / Co-first Author · 2025–2026

Role	Addressed the difficulty of jointly achieving attack effectiveness and human imperceptibility in music retrieval, genre classification, and cover-song identification.
Action	Proposed MAIA: locate critical audio segments and use generative inpainting for white-box and black-box attacks; further proposed the psychoacoustically conditioned PAMT perceptual representation model.
Contribution	MAIA achieved high attack success and audio fidelity across MIR tasks; PAMT reached 0.65 correlation with subjective listening scores and improved robust accuracy by 9.15% on average.

Parameterized Score-to-Audio Expressive Rendering for Creators Suzhou Embodied Intelligence Future Education

Concept Verification Center PoC · Principal Investigator · Sep. 2025–Aug. 2026

Role	Led the Suzhou Embodied Intelligence Future Education Concept Verification Center project as Principal Investigator (PI) , accountable for end-to-end progress, delivery, and validation.
Action	Coordinated expressive-rendering models, macOS application requirements and integration, evaluation design, and user trials; defined import, A/B comparison, auditioning, export, and preset workflows.
Outcome	A single score produces multiple performance styles; the prototype reached TRL 5 in small-scale relevant-setting validation.

Changba Technology Co., Ltd. Audio Processing Intern · Nov. 2020–Jun. 2021

Role	Contributed to singing-voice generation and enhancement research for online karaoke applications, covering audio processing and algorithm validation.
Action	Conducted audio simulation, algorithm testing, and evaluation to support research iteration in singing voice generation and enhancement.
Outcome	Built practical experience from audio algorithms to evaluation in real product settings, providing an engineering foundation for later audio-AI research.

Selected Publications

First / co-first author first · Full list on Google Scholar

ICASSP 2026	Yuxuan Liu et al. <i>Membership Inference Attack Against Music Diffusion Models via Generative Manifold Perturbation</i> . IEEE International Conference on Acoustics, Speech and Signal Processing
ISMIR 2025	Yuxuan Liu et al. <i>MAIA: An Inpainting-Based Approach for Music Adversarial Attacks</i> . International Society for Music Information Retrieval Conference
AAAI Workshop 2026	Yuxuan Liu et al. <i>TS-RaMIA: Membership Inference Attacks for Symbolic Music Generation Models</i> . PMLR 303:1–15. AAAI Conference on Artificial Intelligence Workshop
CMMR 2025	Yuxuan Liu et al. <i>Training a Perceptual Model for Evaluating Auditory Similarity in Music Adversarial Attack</i> . International Symposium on Computer Music Multidisciplinary Research
ICASSP 2026	P Zhang, Yuxuan Liu et al. <i>DDSC: Dynamic Dual-Signal Curriculum for Data-Efficient Acoustic Scene Classification under Domain Shift</i> . IEEE International Conference on Acoustics, Speech and Signal Processing
ICASSP 2026	P Zhang et al. <i>TopSeg: A Multi-Scale Topological Framework for Data-Efficient Heart Sound Segmentation</i> . IEEE International Conference on Acoustics, Speech and Signal Processing
FG 2026	P Zhang, Yuxuan Liu, Z. Li, R. Sang, S. Li. <i>LBOR: Laplace–Beltrami Operator Regularization for Robust Skeleton-based Isolated Sign Language Recognition</i> . pp. 1–5. IEEE International Conference on Automatic Face and Gesture Recognition
DCASE 2025	P Zhang (co-first author), Yuxuan Liu (co-first author) et al. <i>An Entropy-Guided Curriculum Learning Strategy for Data-Efficient Acoustic Scene Classification under Domain Shift</i> . Workshop on Detection and Classification of Acoustic Scenes and Events
AAAI Workshop 2026	R. Sang, Yuxuan Liu. <i>SceneGuard: Training-Time Voice Protection with Scene-Consistent Audible Background Noise</i> . AAAI Conference on Artificial Intelligence Workshop

Core Competencies

Generative

Audio Diffusion models, music/speech generation, symbolic music modeling, and MusicXML workflows

Trustworthy

AI Membership inference, model privacy, copyright auditing, provenance, and adversarial robustness evaluation

Audio

Intelligence MIR, audio signal processing, acoustic scene classification, and perceptually aligned evaluation

Agent

Toolchain Codex, Cursor, and Claude Code for implementation, debugging, research-prototype iteration, and document automation

Engineering

& Research Python, C/C++ , MATLAB, Java, and SQL; experimental design, statistical analysis, academic writing, and user-trial organisation